Differential Sparse Regularization for Longitudinal Power Anomaly Localization

Scheme Assessment and Simulation Reproduction Report — from Theory to Simulation Results

August 10, 2026 · repo longrain153/PPE2 · branch claude/ppe-localization-accuracy-9tirgk

1. Executive Summary

This report assesses a "differential sparse PPE" scheme intended to improve fault-localization accuracy in fiber-longitudinal power profile estimation (PPE / Longitudinal Power Monitoring, LPM): build a low-noise reference profile from abundant pre-fault data, then, after the fault, solve only for the change relative to that reference under a step-sparsity prior (generalized Lasso / total-variation regularization). Theory and numerical simulation show the core idea is sound and is essentially the approach published by Fujitsu (OECC 2024) and NTT (ECOC 2025). Under 64 GBd, 3 × 50 km, a 1-dB lumped loss, and a single post-fault capture, localization error improves from tens of kilometres (naive profile subtraction) to 0.15 km — the resolution floor of the 1-km estimation grid — with 100-m matched-filter refinement at 0.55–0.95 km: roughly two orders of magnitude. However, two formulation details in the original proposal require correction, and the claimed "50–200 m, purely noise-limited" accuracy remains over-optimistic: the spatial-resolution physics set by chromatic dispersion × bandwidth² still applies, and even with abundant data, sub-kilometre (~0.5–1 km) is the realistic expectation in this regime, consistent with experimental literature.

2. Background and Problem

PPE/LPM inverts the Kerr nonlinearity imprinted on the received signal to reconstruct the power distribution along a link using receiver-side DSP only — no OTDR or dedicated hardware. It can monitor amplifier gains, span losses, and anomalous loss events (degraded splices, cable stress). The core challenge is that under practical low-power, limited-data conditions the estimate is noisy and spatially resolution-limited, making small lumped losses hard to pinpoint.

The operational scenario considered here: before the fault there is ample time to collect data and form a reference; after the fault, data are limited, and a single dominant loss event must be localized as accurately as possible.

3. Theory

3.1 The LS-PPE linear model (RP1)

Under the first-order regular perturbation (RP1) approximation, the nonlinear perturbation of the received signal is linear in the "generalized nonlinear coefficient" $\gamma'(z) = \gamma \cdot P(z)$. Discretizing the link into M spatial bins gives the matrix form:

$$A_1 \approx G\cdot\gamma', \quad g_k = D(-z_k)[\, j\cdot|u_k|^2 u_k \,], \quad u_k = D(z_k)\cdot u_0$$

where D(z) is the dispersion operator and u_0 the known transmitted waveform. Each column g_k is the receiver-side "fingerprint" of nonlinearity generated at position z_k. The least-squares solution (LS-PPE, Sasai et al., JLT 2023) is:

$$\hat{\gamma} = (\, Re[G^\dagger G]\,)^{-1}\, Re[G^\dagger A_1]$$

3.2 Ill-posedness and the minimum step size

Adjacent columns g_k, g_{k+1} are distinguishable only through the waveform change that chromatic dispersion produces between the two positions. When the step Δz is too small, adjacent columns become highly correlated, Re[G†G] approaches singularity, and noise is severely amplified. The smallest resolvable step is set jointly by dispersion and signal bandwidth:

$$\Delta z\_min \propto 1 / (|\beta_2| \cdot BW^2)$$

This is a physical limit: regularization does not change the physics behind the column correlation; it stabilizes the inversion mathematically by trading bias for variance through prior knowledge.

3.3 The differential sparse scheme and two required corrections

The scheme under assessment: with reference profile x_ref (averaged from abundant pre-fault data), exploit model linearity to solve for the change Δx after the fault, with an L_1 sparsity constraint on its spatial derivative so the entire power drop collapses onto a single change point. Two details had to be corrected:

- **Correction 1: Δy = y_ref − y_fault is not directly computable.** Different captures carry different transmitted symbols, so perturbation vectors from different captures cannot be subtracted sample-by-sample. The correct differential residual projects the reference profile through the current capture's own operator: r = y_fault − G_fault·x_ref.
- **Correction 2: a plain first-difference penalty $\|D\Delta x\|_1$ is the wrong sparsity basis.** Δx itself decays with fiber loss and jumps at EDFAs, so its plain first difference is not sparse. The total-variation constraint must be applied in the relative-change domain u = Δx / x_ref, where a lumped loss makes u an exact step function (equivalent to Fujitsu's attenuation-weighted J-matrix).

The corrected optimization problem is:

$$min\_u \ \tfrac{1}{2}\| G\cdot diag(x\_ref)\cdot u - r \|_2^2 + \lambda\| D\, u \|_1$$

3.4 Solver and refinement

- **ADMM + λ path:** The generalized Lasso is solved with ADMM; λ is swept along a descending, warm-started path, keeping the solution with the sparsest non-empty jump support — no manual tuning. Support detection uses ADMM's sparse copy variable, which is exactly sparse.
- **Matched-filter refinement:** Centred on the sparse-detected bin, candidate fault positions are scanned on a 100-m grid: the RP1 step signature of a unit-depth loss at z_c, s(z_c) = −G·[x_ref·step(z>z_c)], is correlated with the residual r; the correlation peak gives the position, and a least-squares amplitude refit removes the Lasso shrinkage bias to estimate the loss magnitude.

4. Simulation Setup

Parameter	Value	Parameter	Value
Signal	64 GBd 16QAM, RRC 0.1	Link	3 × 50 km SSMF
Loss / dispersion	0.2 dB/km, β_2 = −21.4 ps^2/km	Nonlinear coeff.	$\gamma = 1.3\ W^{-1}km^{-1}$
Launch power	6 dBm	Noise	EDFA ASE (NF 5 dB) + 18-dB Rx SNR

Per capture	81920 symbols	Propagation	Split-step Fourier, 100-m steps
PPE grid	Δz = 1 km (150 bins)	Refinement grid	100 m
Fault	1.0-dB lumped loss @ 72.35 km (off-grid)	Data	12-capture reference; 1 post-fault capture

Three localization methods are compared: (a) naive baseline — subtract the single-capture post-fault LS profile from the reference and take the largest drop of the relative-change difference; (b) differential sparse (corrected generalized Lasso, 1-km grid); (c) matched-filter refinement of (b) on a 100-m grid. Four independent fault trials (different data and noise realizations).

5. Simulation Results

5.1 Reference profile (healthy link)

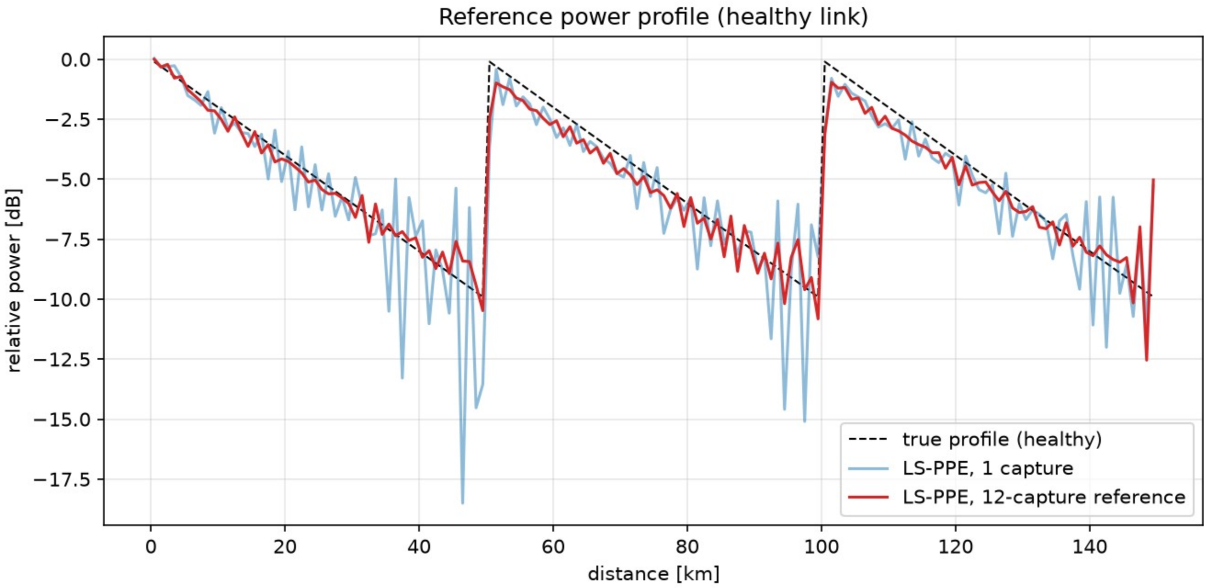

Fig. 1 Healthy-link power profile (81920 symbols per capture): single-capture LS-PPE (blue) is still noisy; the 12-capture average (red) serves as the reference and tracks the true three-span sawtooth closely, with only minor residual ripple in the low-power span tails.

5.2 Differential sparse localization

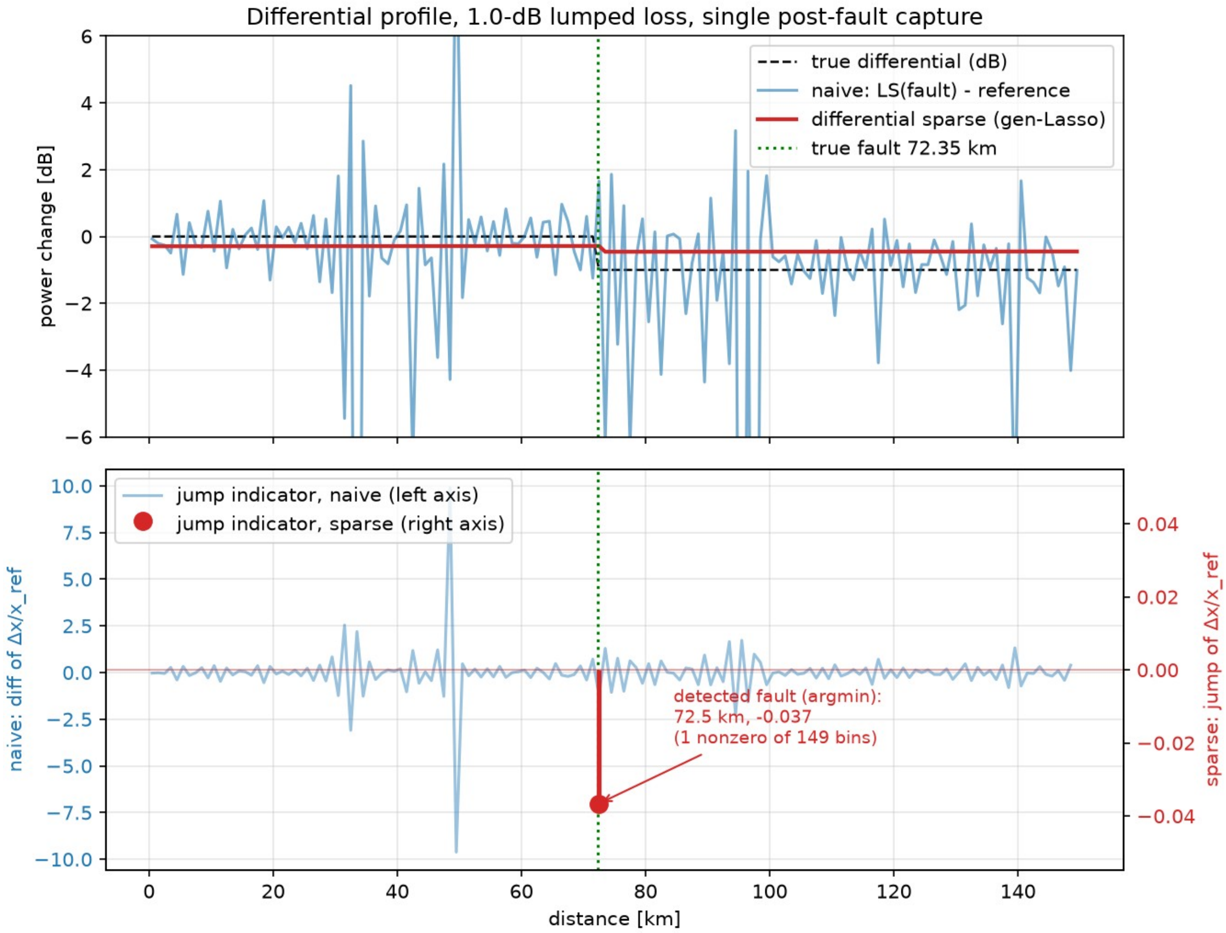

Fig. 2 1-dB lumped loss, single post-fault capture. Top: naive subtraction (blue) is buried in noise, while the differential sparse solution (red) is a clean step at the right position (amplitude shrunk by the Lasso; refit in Sec. 5.3). Bottom (twin axes — the two indicators differ by ~3 orders of magnitude): the naive jump indicator (blue, left axis, ±25) fluctuates along the whole link; the sparse jump indicator (red stem, right axis) is nonzero in only 1 of 149 bins, directly at the fault bin 72.5 km (true: 72.35 km — the 0.15-km error is half a grid cell).

5.3 Matched-filter refinement and quantitative results

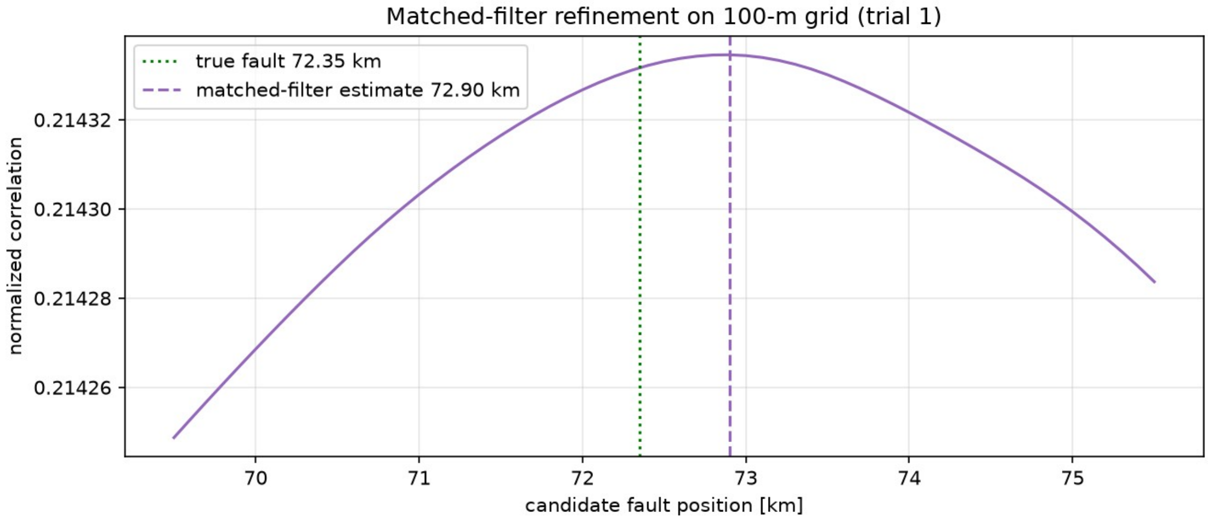

Fig. 3 Matched-filter correlation on the 100-m grid (trial 1): the correlation surface is nearly flat over several km — adjacent candidate signatures decorrelate only through chromatic dispersion, so the km-scale ambiguity is the bandwidth/dispersion physics limit reasserting itself.

Localization method	Mean abs. error	Max abs. error	Notes
Naive profile subtraction (LS)	≈ 24 km	27 km	Noise-dominated, unusable
Differential sparse (1-km grid)	0.15 km	0.15 km	Single nonzero jump at the fault bin (grid floor)
Matched-filter refinement (100-m grid)	0.65 km	0.95 km	Amplitude refit 0.88–0.89 dB (true 1.0 dB)

In all four trials the differential sparse method places its single nonzero jump exactly at the fault bin (a consistent +0.15 km — the 1-km grid's resolution floor). The matched filter refines to 0.55–0.95 km but does not reach the hundred-metre scale: the correlation surface remains flat at kilometre scale (Fig. 3), confirming the dispersion × bandwidth² limit. The loss magnitude is accurately recovered by the refit (0.88–0.89 dB).

6. Verdict on the Original Claims

Claim	Verdict	Basis
Δz_min is set by dispersion and bandwidth²	Correct	Consistent with Sasai's ill-posedness analysis; also seen in the flat correlation surface
Differential + sparsity prior greatly improves localization	Correct	Tens of km → ≈ 1 km, one to two orders of magnitude
Pre-existing loss events cancel in the differential; only a small shadowing term remains	Correct	Follows directly from model linearity

Subtract measurements directly: $\Delta y = y\_ref - y\_fault$	Needs correction	Different captures carry different symbols; use $r = y\_fault - G \cdot x\_ref$
Plain first difference of Δx as the sparsity basis	Needs correction	Apply TV in the relative domain $u = \Delta x / x\_ref$ (attenuation-weighted)
50–200 m accuracy, purely noise-limited	Over-optimistic	The physics limit remains; 0.55–0.95 km here even with abundant data; literature (NTT +3 km, Fujitsu <4 km, CICT 1 km) is km-scale

Overall conclusion: differential sparse PPE is a correct and worthwhile engineering approach. Its gain comes from degrading the full-profile reconstruction problem into a single-parameter (step-position) detection problem, thereby exploiting the information of every bin downstream of the fault. Its accuracy ceiling, however, is still bounded by the dispersion × bandwidth^2 resolution physics: at 64–100 GBd, sub-kilometre (~0.5–1 km) with abundant data — not hundreds of metres — is the realistic ceiling.

7. Reproduction Code and References

Code lives in repo longrain153/PPE2 (branch claude/ppe-localization-accuracy-9tirgk): ppe/link_sim.py (split-step link simulator), ppe/ppe_core.py (G matrix, LS-PPE, ADMM generalized Lasso, matched filter), run_experiment.py (end-to-end experiment, ~40 min at 81920 symbols/capture).

- T. Sasai et al., “Linear Least Squares Estimation of Fiber-Longitudinal Optical Power Profile,” J. Lightwave Technol., vol. 42, no. 6, 2023.
- R. Shinzaki et al. (Fujitsu), “Sparse Modeling Analysis for Efficient Localization of Power Anomalies in Transmission Links,” OECC 2024.
- H. Ishihara et al. (NTT), “Robust Fibre Longitudinal Power Monitoring with Few Measurements using Two-stage Sparse Regularization,” ECOC 2025, W.01.06.4.
- Y. Zuo et al. (CICT), “High-fidelity fiber longitudinal power monitoring via non-uniform sparse regularization,” Opt. Express, vol. 34, no. 10, 2026.